Bank profitability and the business cycle

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Abstract

An important element of the macro-prudential analysis is the study of the link between business cycle fluctuations and banking sector profitability and how this link is affected by institutional and structural characteristics. This work estimates a set of equations for net interest income, non-interest income, operating costs, provisions, and profit before taxes, for banks in the main industrialized countries and evaluates the effects on banking profitability of shocks to both macroeconomic and financial factors. Distinguishing mainly the euro area from Anglo-Saxon countries, the analysis also identifies differences in the resilience of the respective banking systems and relates them to the characteristics of their financial structure.

Introduction

The knowledge of the link between business cycle fluctuations and banking sector profitability is important in order to evaluate the stability and soundness of the financial and banking sector. Bad economic conditions can worsen the quality of the loan portfolio, generating credit losses, which eventually reduce banks' profits. Depending on their capitalization, the capacity of banks to sustain the activity of the private sector may also be jeopardized, and the fluctuations of the business cycle may be exacerbated (Gambacorta and Mistrulli, 2004). Structural factors are important in studying the link between bank profitability and business cycle fluctuations; relevant examples are the presence of lending relationships, the level of competition in the local credit markets, and the development of the stock and capital markets.

These considerations are at the basis of a different approach to the issue of financial stability which is often referred to as macro-prudential analysis. The latter is an important tool for bank regulators as it supports micro-prudential supervision by evaluating the overall robustness of the macroeconomic environment and by detecting early signals of financial distress (Kaminsky, 1999, Logan, 2000, Borio, 2003). This is not achieved by trying to anticipate the shock that triggers a “crisis”, but rather by detecting situations of “fragility” that involve the structure of the financial system (Davis, 1999, Bell and Pain, 2000, IMF, 2001).

Empirical findings show that bank profitability is an important predictor of financial crises (Demirgüç-Kunt and Detragiache, 1999). However, the monitoring of bank profits is made difficult by the fact that bank profit components are observed only at low frequencies, at best quarterly; detailed public information is available only for large and listed companies. Accordingly, studying how macroeconomic and structural indicators influence banks’ profits is important as such indicators are observed with higher frequency (especially those on the financial markets).

Fluctuations of bank profitability matter also because of the “bank capital channel” (Van den Heuvel, 2002), which is based on the hypothesis of an imperfect market for bank equity: banks cannot easily issue new equity because of the presence of agency costs and tax disadvantages (Myers and Majluf, 1984, Cornett and Tehranian, 1994, Calomiris and Hubbard, 1995, Stein, 1998). The mechanism is the following: after a drop in bank profitability, if equity is sufficiently low and it is too costly to issue new shares, banks reduce lending, otherwise they fail to meet regulatory capital requirements and this produces real effects on consumption and investment.[1]

Building an econometric model of the link between business cycle indicators and the main components of bank balance sheets is important in the light of the development of the Financial Sector Assessment Program (FSAP), jointly established by the International Monetary Fund (IMF) and the World Bank in 1999, which specifically looks at countries’ financial sectors, assessing strengths and vulnerabilities in order to reduce the potential for crisis (Hoggarth, 2003, Calari and Ingves, 2005). A more recent review of this programme, in fact, indicates that the assessment of financial sector vulnerability is not presently based upon an explicitly tested model (Kupiec, 2005).

This paper studies the link between bank profitability and the business cycle by using data for 10 industrialized countries (Austria, Belgium, France, Germany, Italy, the Netherlands, Portugal, Spain, United Kingdom and United States) over the period 1981–2003. The dataset includes yearly figures from the balance sheet and the income statement of the aggregated national banking industries, collected by OECD in a harmonized way that minimizes the effects of differences in accounting and statistical definitions and allows meaningful comparisons across countries.[2] The main novelty of this paper lies in a comprehensive analysis of the effects of the business cycle on all income statement components (net interest income, non-interest income, operating expenses, and provisions).

The results can be summarized as follows. Since the mid-1980s the levels of profitability of banking systems in the euro area have registered a wide convergence, although to a lower level compared to US and UK. The regression analysis shows that bank profitability in Anglo-Saxon economies has been structurally higher, even taking into account the differences in economic cycles, in the development in financial systems and in taxation. On the contrary, the evidence suggests that the higher level of bank profitability in UK and US is at least partly related to their more flexible cost structure.

We find that bank profits pro-cyclicality derives from the effect that the economic cycle exerts on net interest income (via lending activity) and loan loss provisions (via credit portfolio quality). Non-interest income is not significantly influenced by GDP changes suggesting that, contrary to previous findings, revenue diversification could contribute to the stabilization of bank profitability. Financial deepness stimulates bank profitability.

The remainder of the paper is organized as follows. Section 2 analyzes macroeconomic and structural developments in each country in the panel after the introduction of the euro. Section 3 describes the econometric model and the results for each income statement component, the overall profit before taxes and return on equity (ROE). Section 4 analyzes differences between euro-area and Anglo-Saxon countries in terms of the level of profitability, reaction of bank profit to GDP and interest rate shocks and sluggishness of operating costs. The last section summarizes the main conclusions.

Section snippets

Some facts about bank profitability and the business cycle

Since the mid-1980s, dramatic changes in regulation, demand composition and technology have modified the structure and the boundaries of credit markets (Bhattacharya et al., 1998). All these changes have strengthened competition, especially in traditional lending activity, reduced intermediation margins and encouraged banks to diversify their sources of revenue and increase efficiency in production and distribution.

Since the mid-1990s cross-country variability of gross income as a percentage of

The empirical evidence

Indexing countries with j and years with t, the econometric analysis is carried out using the following benchmark model:where Y_{jt} is the income statement component examined, X_{jt} is a vector of explanatory variables, T_t is a vector of year-dummies, η_j is an unobservable time-invariant country effect. In particular, X_{jt} = [GDP_{jt}, $DCPI_{jt}$, MMR_{jt}, LTP_{jt}, SMC_{jt}, BL_{jt}, VSM_{jt}, TA_{jt}] where GDP_{jt} is the level of real gross domestic product, $DCPI_{jt}$ is the rate of

Differences between periods and groups of countries

The aim of this section is twofold. First, we wish to explore in greater detail whether the introduction of the euro determined structural breaks in each income statement component. Second, we hope to shed some light on the existence of differences across the euro-area and Anglo-Saxon countries both in the level of profitability and in its sensitivity to macroeconomic and structural indicators.[9]

Conclusions

This work estimates a set of equations for net interest income, non-interest income, operating costs, provisions, and profit before taxes for banks in the main industrialized countries, in order to evaluate the effects on banking profitability of shocks to macroeconomic and financial factors. The main results are the following.

The dispersion of bank profitability among euro-area countries has declined dramatically since the mid-1990s, during the convergence process towards the third stage of

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